

Week4_Yuliia_Chernysheva

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Instructions

Using the file mbox-short.txt as the source file.

Write a program that:

- Prompts for a filename
- Then opens that file and reads through it, looking for lines like X-DSPAM-Confidence
- Extract the floating-point values from the end of each of the lines
- Sum the total
- Count how many lines
- Compute the average of those values and produce output like:
Count: 27
Mean is: 0.7507185185185187

```
[1]: from pathlib import Path

# Constant for searching
TARGET_PHRASE = "X-DSPAM-Confidence"

# Prompts for a filename
file_name = input("Enter a file name >> ").strip()
file_path = Path(file_name)

# Check for the existence of the file
if file_path.exists():
    total = 0
    count = 0
    # Open the file for reading
    with file_path.open("r", encoding="utf-8") as f:
        for line in f:
            if line.startswith(TARGET_PHRASE):
                # Find the position of the colon
                colon_pos = line.find(':')
                # extract the part after the colon
                value = line[colon_pos + 1:].strip()
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        # Updating the counters
        total += float(value)
        count += 1

    # Output the results
    print(f"Count = {count}")
    if count > 0:
        # Calculate the mean
        mean_value = total / count
        print(f"Mean is {mean_value:.5f}")
    else:
        print("Mean is: no data to calculate average")
else:
    print(f"Error: File '{file_name}' not found")
```

Enter a file name >> mbox-short.txt

Count = 27

Mean is 0.75072

[]: